A First Course in Abstract Algebra

Section 54: Illustrations of Galois Theory

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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54 Illustrations of Galois Theory

Exercise 54.1: Irreducibility of the eighth cyclotomic polynomial

Make the invertible substitution $x = y + 1$. Then

$$(y+1)^4 + 1 = y^4 + 4y^3 + 6y^2 + 4y + 2.$$

Eisenstein's criterion at 2 applies: every coefficient except the leading one is divisible by 2, and the constant 2 is not divisible by 4. Thus the substituted polynomial is irreducible over $\mathbb{Q}$. A factorization of $x^4 + 1$ would yield a factorization after substitution, so $x^4 + 1$ is irreducible too.

Exercise 54.2: Checking the entire octic fixed-field diagram

Put $\alpha = \sqrt[4]{2}$ and $K = \mathbb{Q}(\alpha, i)$. We have $[K : \mathbb{Q}] = 8$, since $x^4 - 2$ is Eisenstein and $\mathbb{Q}(\alpha)$ is real. To avoid relying on a picture alone, write $r(\alpha) = i\alpha, r(i) = i$ and $s(\alpha) = \alpha, s(i) = -i$. Then $r^4 = s^2 = 1$ and $srs = r^{-1}$. In the source notation $r = \rho_1$, $r^2 = \rho_2$, $s = \mu_1$, $r^2 s = \mu_2$, $rs = \delta_1$, and $r^3 s = \delta_2$. The proposed fields can all be verified by their fixed generators and degrees:

subgroup	proposed fixed field	degree over $\mathbb{Q}$
$\langle r^2, s\rangle = H_1$	$\mathbb{Q}(\alpha^2) = \mathbb{Q}(\sqrt{2})$	2
$\langle r\rangle = H_2$	$\mathbb{Q}(i)$	2
$\langle r^2, rs\rangle = H_3$	$\mathbb{Q}(i\alpha^2) = \mathbb{Q}(\sqrt{-2})$	2
$\langle s\rangle = H_4$	$\mathbb{Q}(\alpha)$	4
$\langle r^2 s\rangle = H_5$	$\mathbb{Q}(i\alpha)$	4
$\langle r^2\rangle = H_6$	$\mathbb{Q}(\alpha^2, i)$	4
$\langle rs\rangle = H_7$	$\mathbb{Q}((1+i)\alpha)$	4
$\langle r^3 s\rangle = H_8$	$\mathbb{Q}((1-i)\alpha)$	4

For each row, substitution of r, s checks that the indicated generator is fixed by the subgroup. For H_1, H_2, H_3, the generator is a nonrational quadratic radical, giving degree 2. For H_4, H_5, the fourth power is 2 and $x^4 - 2$ is irreducible. For H_6, the real quadratic field gains the nonreal i, giving degree 4. For H_7, H_8, if $\gamma = (1 \pm i)\alpha$, then $\gamma^2 = \pm 2i\sqrt{2}$ and $\gamma^4 = -8$. Its square generates $\mathbb{Q}(\sqrt{-2})$, and $\pm 2i\sqrt{2}$ is not a square there: writing $(a + b\sqrt{-2})^2 = \pm 2\sqrt{-2}$ would force $a^2 - 2b^2 = 0$ and $ab = \pm 1$, impossible with rational a, b. Thus these fields have degree 4 too. In every row this degree equals the subgroup index, proving equality with the full fixed field. The trivial subgroup fixes K; the full group fixes $\mathbb{Q}$. This verifies all ten fields and, by reversing subgroup containment, all inclusions in the diagram.

Exercise 54.3: Primitive generators and all their minimal polynomials

Use the notation of Exercise 54.2. A complete list, including top and bottom, is

field	primitive element	minimal polynomial
$\mathbb{Q}$	0	x
$\mathbb{Q}(\sqrt{2})$	$\sqrt{2}$	x^2-2
$\mathbb{Q}(i)$	i	x^2+1
$\mathbb{Q}(\sqrt{-2})$	$i\sqrt{2}$	x^2+2
$\mathbb{Q}(\alpha)$	α	x^4-2
$\mathbb{Q}(i\alpha)$	$i\alpha$	x^4-2
$\mathbb{Q}(\sqrt{2},i)$	$\sqrt{2}+i$	x^4-2x^2+9
$\mathbb{Q}((1+i)\alpha)$	$(1+i)\alpha$	x^4+8
$\mathbb{Q}((1-i)\alpha)$	$(1-i)\alpha$	x^4+8
K	$\alpha+i$	$x^8+4x^6+2x^4+28x^2+1$

The degrees and irreducibility in the first nine rows follow from Exercise 54.2 and the four distinct sign conjugates of $\sqrt{2}+i$. For the last row, the eight images of $\alpha+i$ are $i^j\alpha+\varepsilon i$ for $0\le j<4$ and $\varepsilon=\pm1$. They are all distinct: an equality with different j would express α as an element of $\mathbb{Q}(i)$, impossible because its rational degree is 4; with equal j the signs must agree. Thus this element has degree 8 and generates K. Multiplying

$$((x-i)^4-2)((x+i)^4-2)$$

gives the displayed monic polynomial, proving it is its minimal polynomial. The identical quartics in several rows do not imply identical subfields: an abstract isomorphism can send one primitive root to another.

Exercise 54.4: The fifth-root-of-unity correspondence

Let ζ be a primitive fifth root.

(a) Its powers $1,\zeta,\zeta^2,\zeta^3,\zeta^4$ are all roots of x^5-1, so $K=\mathbb{Q}(\zeta)$ is exactly its splitting field.

(b) An automorphism preserves order and the equation $\zeta^5=1$, so sends ζ to ζ^a for $a=1,2,3,4$.

(c) Φ_5 is irreducible of degree 4. Thus every listed image gives an automorphism, $\sigma_a(\zeta)=\zeta^a$. Composition multiplies exponents modulo 5. Since 2 has order 4 modulo 5, the group is $\langle\sigma_2\rangle\cong\mathbb{Z}_4$.

(d) Its only proper nontrivial subgroup is $\langle\sigma_4\rangle$, generated by complex conjugation. Its fixed field is $M=\mathbb{Q}(t)$ where $t=\zeta+\zeta^{-1}$. Dividing $1+\zeta+\cdots+\zeta^4=0$ by ζ^2 gives $t^2+t-1=0$, so $M=\mathbb{Q}(\sqrt{5})$ has degree 2 and is the full fixed field. The two diagrams are the chains

$$\begin{array}{cc} K & \langle\sigma_2\rangle \\ | & | \\ M & \langle\sigma_4\rangle \\ | & | \\ \mathbb{Q} & \{1\} \end{array}$$

with correspondence $K\leftrightarrow\{1\}$, $M\leftrightarrow\langle\sigma_4\rangle$, and $\mathbb{Q}\leftrightarrow\langle\sigma_2\rangle$. Each step has index or field degree 2.

Exercise 54.5: A fifth radical over a cyclotomic base

Put $F = \mathbb{Q}(\zeta_5)$ and $r = \sqrt[5]{2}$. Since $[F : \mathbb{Q}] = 4$ and $[\mathbb{Q}(r) : \mathbb{Q}] = 5$, the tower law applied to $F(r)$ shows $5 \mid 4[F(r) : F]$. But adjoining a root of $x^5 - 2$ gives degree at most 5, so $[F(r) : F] = 5$. The presence of ζ_5 makes all roots $r\zeta_5^j$ lie in $F(r)$, so this is the splitting field. The map $r \mapsto r\zeta_5$, fixing F, has order 5 and generates all five automorphisms. Thus the group is $\mathbb{Z}_5$.

Exercise 54.6: The seventh-root-of-unity correspondence

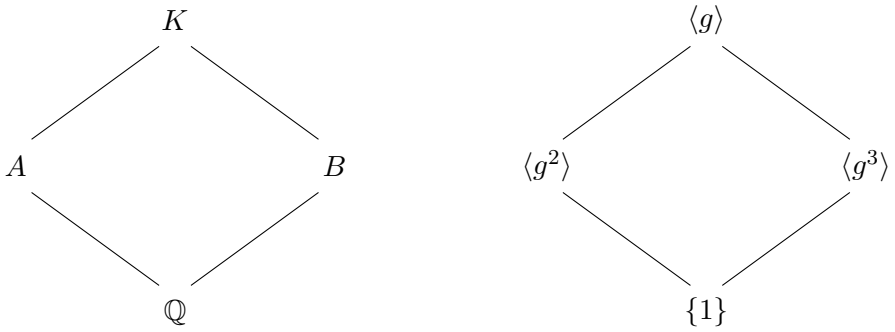
Let ζ have order 7 and $K = \mathbb{Q}(\zeta)$.

(a) Its seven powers are precisely the roots of $x^7 - 1$, so it generates the splitting field.

(b) Every automorphism sends ζ to one of ζ^a, $1 \le a \le 6$.

(c) Irreducibility of Φ_7 gives degree 6, so all six exponent maps occur. Their group is $(\mathbb{Z}/7\mathbb{Z})^\times$, cyclic of order 6 generated by 3. Write $g = \sigma_3$.

(d) The two proper nontrivial subgroups are $\langle g^2 \rangle = \{\sigma_1, \sigma_2, \sigma_4\}$ of order 3 and $\langle g^3 \rangle = \{\sigma_1, \sigma_6\}$ of order 2. Their fixed fields are respectively $A = \mathbb{Q}(\sqrt{-7})$ and $B = \mathbb{Q}(\zeta + \zeta^{-1})$. For the first, put $\eta = \zeta + \zeta^2 + \zeta^4$. The other three powers sum to $\bar{\eta}$, and direct multiplication gives $\eta + \bar{\eta} = -1$, $\eta\bar{\eta} = 2$. Thus η satisfies $x^2 + x + 2$, has degree 2, and generates A. The subgroup fixes this sum, so it fixes exactly A. For B, complex conjugation fixes $t = \zeta + \zeta^{-1}$ and ζ satisfies $X^2 - tX + 1$; since B is real and ζ nonreal, $[K : B] = 2$ and $[B : \mathbb{Q}] = 3$. For a check, $t^3 + t^2 - 2t - 1 = 0$ follows by dividing $\Phi_7(\zeta) = 0$ by ζ^3. The field and group diagrams are diamonds:



The correspondence reverses order: $A \leftrightarrow \langle g^2 \rangle$, $B \leftrightarrow \langle g^3 \rangle$, with top and bottom paired oppositely. There are no other intermediate fields because a cyclic group of order 6 has no other subgroups.

Exercise 54.7: Eighth roots without recomputing a table

The splitting field is $\mathbb{Q}(\zeta_8) = \mathbb{Q}(\sqrt{2}, i)$. Its group is $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$ under multiplication modulo 8. Each of its three nonidentity elements squares to 1, so it is the Klein four-group $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Exercise 54.8: A complete quartic splitting-field correspondence

Find the field and its degree.

Put $a = \sqrt{2+\sqrt{5}}$, $b = i/a$, and $K = \mathbb{Q}(a, i)$. Then $b^2 = 2 - \sqrt{5}$ and $ab = i$, so the roots are $a, -a, b, -b$ and K is their splitting field. The norm of $2 + \sqrt{5}$ from $\mathbb{Q}(\sqrt{5})$ to $\mathbb{Q}$ is -1, so it is not a square there. Thus $[\mathbb{Q}(a) : \mathbb{Q}] = 4$; this field is real and gains degree 2 on adjoining i, so $[K : \mathbb{Q}] = 8$.

Describe the group.

Define

$$r(a) = b, \quad r(i) = -i, \qquad s(a) = a, \quad s(i) = -i.$$

To justify existence, conjugacy gives an embedding $\mathbb{Q}(a) \to \mathbb{Q}(b)$ sending a to b. The element i is outside $\mathbb{Q}(b)$: otherwise $a = i/b$ would also belong to that degree-four field, contradicting $[K : \mathbb{Q}] = 8$. Thus $x^2 + 1$ is still irreducible over $\mathbb{Q}(b)$, and the embedding extends with either choice $i \mapsto \pm i$. The map s is ordinary complex conjugation. This constructs the asserted automorphisms, rather than merely proposing images. They satisfy $r^4 = s^2 = 1$, $srs = r^{-1}$, and yield eight distinct maps $1, r, r^2, r^3, s, rs, r^2s, r^3s$. Thus the group is the dihedral group of order 8.

List every subgroup and its field.

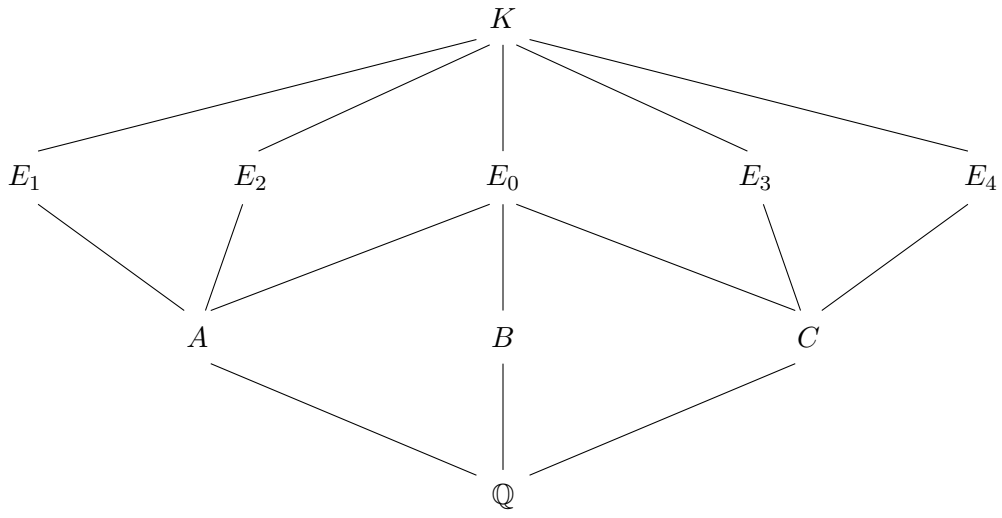
Define the field labels by this table:

subgroup	fixed field	[field : $\mathbb{Q}$]
G	$\mathbb{Q}$	1
$\langle r \rangle$	$B = \mathbb{Q}(i\sqrt{5})$	2
$\langle r^2, s \rangle$	$A = \mathbb{Q}(\sqrt{5})$	2
$\langle r^2, rs \rangle$	$C = \mathbb{Q}(i)$	2
$\langle r^2 \rangle$	$E_0 = \mathbb{Q}(\sqrt{5}, i)$	4
$\langle s \rangle$	$E_1 = \mathbb{Q}(a)$	4
$\langle r^2 s \rangle$	$E_2 = \mathbb{Q}(ia)$	4
$\langle rs \rangle$	$E_3 = \mathbb{Q}(a + b)$	4
$\langle r^3 s \rangle$	$E_4 = \mathbb{Q}(a - b)$	4
$\{1\}$	K	8

For verification, r negates both i and $\sqrt{5} = a^2 - 2$, hence fixes $i\sqrt{5}$. The subgroup $\langle r^2, s \rangle$ fixes a^2, and $\langle r^2, rs \rangle$ fixes i. The individual involutions fix their displayed generators: r^2 fixes a^2, i; s fixes a; r^2s fixes ia; rs interchanges a, b; r^3s interchanges $a, -b$. The quadratic degrees are immediate. The field $E_1 = \mathbb{Q}(a)$ has degree 4 as proved above. For E_2, the element ia satisfies $X^4 + 4X^2 - 1 = 0$. Its images under the already constructed automorphisms include $ia, -ia, ib, -ib$, four distinct numbers: $\pm ia$ are nonzero imaginary and $\pm ib = \mp 1/a$ are nonzero real. Its minimal polynomial therefore has at least four distinct roots and degree at most 4, so its degree is exactly 4. For E_0, i doubles the real quadratic degree. Finally $(a \pm b)^2 = 4 \pm 2i$, which is not a square in $\mathbb{Q}(i)$: the norm of a putative square would be the rational square 20, impossible. Hence E_3, E_4 have degree 4, with minimal polynomial $x^4 - 8x^2 + 20$. All proposed fields therefore have the required index and equal the full fixed fields.

Give all inclusions.

The following is the full field diagram. Reverse its edges and replace each field by its subgroup from the table to obtain the full subgroup diagram. In particular E_0 contains all three quadratic fields; E_1, E_2 contain only A, and E_3, E_4 contain only C.



To make the subgroup diagram explicit as well, its covering relations are

upper subgroup	immediately lower subgroups
G	$\langle r\rangle,\ \langle r^2, s\rangle,\ \langle r^2, rs\rangle$
$\langle r\rangle$	$\langle r^2\rangle$
$\langle r^2, s\rangle$	$\langle r^2\rangle, \langle s\rangle, \langle r^2 s\rangle$
$\langle r^2, rs\rangle$	$\langle r^2\rangle, \langle rs\rangle, \langle r^3 s\rangle$
each order-two subgroup	$\{1\}$

These relations and the field diagram exhibit the entire correspondence, not just its normal subgroups.

Exercise 54.9: Two symmetric-function identities

Write $s_1 = y_1 + y_2 + y_3$, $s_2 = y_1y_2 + y_1y_3 + y_2y_3$, $s_3 = y_1y_2y_3$.

(a) Expanding s_1^2 gives $y_1^2 + y_2^2 + y_3^2 + 2s_2$, so the answer is $s_1^2 - 2s_2$.

(b) Put the six displayed ratios over the denominator s_3. Their numerator is $\sum_{i\neq j} y_i^2 y_j$. On expanding s_1s_2, this sum appears together with $3y_1y_2y_3$. Thus the answer is $(s_1s_2 - 3s_3)/s_3$. The equality is an identity of rational functions, and numerical evaluation requires each original denominator nonzero.

Exercise 54.10: Polynomials of a sum and of squares

Vieta's identities for $x^3 - 4x^2 + 6x - 2$ give $s_1 = 4, s_2 = 6, s_3 = 2$ for its three roots.

(a) Their sum is 4, so the monic polynomial with that sole zero is $x - 4$.

(b) The sum of the squared roots is $s_1^2 - 2s_2 = 4$. Their pairwise products sum to $s_2^2 - 2s_1s_3 = 36 - 16 = 20$, and their product is $s_3^2 = 4$. Hence the required polynomial is $x^3 - 4x^2 + 20x - 4$. These squared roots are distinct: an opposite pair of original roots would force the third root to be 4, but the original polynomial does not vanish at 4.

Exercise 54.11: Every finite group occurs over a suitable field

Let H be finite. By Cayley's theorem embed it into S_n for some n. Take independent indeterminates $y_1, \ldots, y_n$ over $\mathbb{Q}$, and put $L = \mathbb{Q}(y_1, \ldots, y_n)$ and $B = \mathbb{Q}(s_1, \ldots, s_n)$, using their elementary symmetric functions. The symmetric-function theorem gives $G(L/B) \cong S_n$. Regard H as a subgroup and let $F = L_H$. The Galois correspondence gives $G(L/F) = H$, and L/F is a finite normal extension. This proves existence over a suitable field F; it does not assert that every finite group occurs over $\mathbb{Q}$.

Exercise 54.12: Discriminants and even permutations

Let $\Delta = \prod_{i<j}(\alpha_i - \alpha_j)$, with roots repeated according to the multiplicities in f.

(a) $\Delta = 0$ exactly when two entries in this root list coincide, meaning f has a repeated root. Since each irreducible factor is separable, repetitions occur exactly when some irreducible factor appears at least twice. This is the stated square-divisibility criterion.

(b) An automorphism permutes the root list and sends Δ to $\pm\Delta$. Squaring removes the sign, so Δ^2 is fixed by the full Galois group and belongs to F.

(c) The equivalence needs two additional hypotheses: f must have distinct roots and $\operatorname{char} F \neq 2$. Under those hypotheses the root action embeds $G(K/F)$ into S_n, and

$$\sigma(\Delta) = \operatorname{sgn}(\sigma)\Delta.$$

This sign rule follows because interchanging two roots negates the alternating product and transpositions generate S_n. Since $\Delta \neq 0$ and $1 \neq -1$, it is fixed exactly by even permutations. Thus $G(K/F) \subseteq A_n$ if and only if every automorphism fixes Δ, equivalently $\Delta \in F$. As printed, the unrestricted assertion is false. In characteristic 2, x^2+x+1 over $\mathbb{Z}_2$ has splitting field of order 4 and group S_2, but $\Delta = \alpha - \alpha^2 = 1 \in \mathbb{Z}_2$. With repeated roots, $\Delta = 0 \in F$ regardless of the group, and the set of distinct roots may have fewer than n elements, so the stated identification with permutations of n distinct roots is not available. The preceding corrected version is the usual discriminant test.

Exercise 54.13: Algebraic integers form a ring

Let α, β satisfy monic integer polynomials of degrees m, n. Higher powers of either can be reduced using its monic relation with integer coefficients. Thus the additive group

$$M = \sum_{0 \le j < m,\ 0 \le k < n} \mathbb{Z}\alpha^j \beta^k$$

is finitely generated, contains 1, and is closed under multiplication by α and β, hence by $\alpha + \beta$, $\alpha\beta$, and $-\alpha$. It is torsion free as a subgroup of the additive group of $\mathbb{C}$, so it is a free abelian group of finite rank. For each γ among these three elements, multiplication by γ on M is represented in a $\mathbb{Z}$-basis by an integer matrix A. Its characteristic polynomial is monic with integer coefficients, and Cayley–Hamilton gives $p(A) = 0$. Applying that identity to $1 \in M$ yields $p(\gamma) = 0$ in $\mathbb{C}$. Thus $\alpha + \beta$, $\alpha\beta$, and $-\alpha$ are algebraic integers. Finally 0 and 1 satisfy x and $x - 1$. These facts verify all subring conditions.